Kinarm DDM report

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Contents

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Bottom line

A hierarchical drift-diffusion model is a defensible starting point for the Kinarm task if the modeled event is defined narrowly as the **initial commitment to launch action** rather than the full reach trajectory or later online corrections (Barany et al., 2020; Haith et al., 2016; Ratcliff & McKoon, 2008). The literature supports a stronger justification for this narrow event than for a broad claim that the whole interception sequence is generated by a standard DDM.

The core asymmetry in the literature is clear. **Drift rate** and **boundary separation** are useful latent parameters, but their absolute values are strongly shaped by scaling, task structure, and model specification (Ratcliff et al., 2016; Tran et al., 2020). **Nondecision time** can be constrained more directly by task timing and physiology, but it is also the parameter most likely to absorb overlap between deciding, preparing, and executing movement if left weakly constrained (Bompas et al., 2024; Weindel et al., 2020).

Why a hierarchical DDM fits the problem

The Kinarm task is trial-rich, participant-variable, and likely heterogeneous across stationary and moving-target conditions. Hierarchical diffusion models are well suited to this setting because they estimate participant-level and group-level parameters together, with shrinkage stabilizing noisy individual estimates (Vandekerckhove et al., 2011). HDDM-style implementations make that practical while also allowing informative priors that keep weakly identified parameters in plausible regions (Wiecki et al., 2013).

A second reason to prefer a hierarchical formulation is that several relevant quantities are hard to estimate cleanly in single subjects or low-trial conditions, especially nondecision time and any across-trial variability terms (Böhm et al., 2018; Wiecki et al., 2013). Partial pooling therefore does real methodological work here rather than merely improving convenience.

What each focal parameter should mean in this project

Parameter	Working interpretation	Main caution
Drift rate v	Efficiency and direction of evidence accumulation toward an initial action choice	Absolute values are not portable across scaling conventions or task families (Ratcliff et al., 2016; Tran et al., 2020)
Boundary separation a	Commitment criterion or response caution before movement initiation	It is a policy term, not a physiological latency bound (Heitz & Schall, 2012; Ratcliff & McKoon, 2008)

Parameter	Working interpretation	Main caution
Nondecision time T_{er}	Time outside the modeled accumulation process, including encoding, preparation, and output stages not assigned to the decision variable	In visuomotor tasks it is often a mixed quantity rather than a pure sensorimotor delay (Bompas et al., 2024; Weindel et al., 2020)

What can be justified directly from the DDM literature

The strongest generic source for priors and plausibility envelopes is the systematic review in (Tran et al., 2020). That review pooled published DDM fits, rescaled accumulation parameters to the $s = 1$ convention, and proposed informative univariate priors.

Parameter	Review-based empirical envelope at $s = 1$	How it should be used here
Drift rate v	About 0.01 to 18.51 in magnitude (Tran et al., 2020)	Outer empirical envelope only, not a task-specific center
Boundary separation a	About 0.11 to 7.47 (Tran et al., 2020)	Plausibility ceiling and prior source, not a literal human limit
Nondecision time T_{er}	About 0 to 3.69 s (Tran et al., 2020)	Too broad to use directly for the Kinarm task

That review also gives the key scaling warning: if the implementation fixes within-trial noise at $s = 0.1$ rather than $s = 1$, then comparable values for v and a are ten times smaller, while time parameters are unchanged (Ratcliff et al., 2016; Tran et al., 2020).

The standard DDM review in (Ratcliff & McKoon, 2008) supplies the other core justification: a DDM is meant for relatively fast, single-stage choices and treats observed RT as decision time plus nondecision time. That logic fits the Kinarm task only if the modeled response is the first commitment to move, not the entire movement with its later adjustments.

What the reaching and eye-hand literature adds

Hand reaction time contains both preparation and initiation delay

In a visually guided reaching task, free reaction time averaged about 212 ms, while movement preparation was inferred at about 130 ms, leaving an additional initiation delay of about 78 ms (Haith et al., 2016). This is one of the clearest task-facing constraints in the literature. It implies that very small fitted hand T_{er} values would be hard to defend, and it also shows that not all observed delay before movement onset is evidence accumulation.

Coordinated eye-hand behavior often shares a decision stage and then diverges

A common stochastic accumulator with effector-specific delay explained coordinated eye-hand RTs well in (Gopal et al., 2015). The same line of work also shows that the architecture is task-dependent: some contexts favor common eye-hand accumulation, while others favor more separate or weakly coupled processes (Jana et al., 2017). For the present project, that makes a shared decision stage with separate eye and hand output terms more plausible than a model that forces both effectors into one common nondecision term.

Interception increases urgency and overlap between deciding and acting

The closest task analogue in the search is (Barany et al., 2020), where participants used a robotic manipulandum for reaching and interception. Interception produced smaller RT increases under added perceptual demand, more decision errors, and more online redirection than stationary reaching. The main implication is that moving-target trials encourage earlier launch and greater overlap between decision formation and movement execution.

This is where the Kinarm report should widen its lens slightly. A standard hierarchical DDM remains a reasonable baseline model, but fitted parameters from moving-target trials should be interpreted as compact summaries of a richer control process rather than literal isolated neural stages (Barany et al., 2020; Dendauw et al., 2024; Selen et al., 2012).

Nondecision time is useful but not literal

The sharpest caution comes from (Bompas et al., 2024), which argues that fitted nondecision time often acts as a mathematical sink for what the chosen decision model fails to capture. The paper distinguishes biologically grounded visuomotor dead time from model-estimated T_{er}, and shows that the two can diverge substantially.

EMG-based decomposition reaches a similar conclusion. In (Weindel et al., 2020), motor time changed with speed–accuracy stress, stimulus strength, and correctness, which means processes usually assigned to T_{er} are not sealed off from decisional manipulations. In short, T_{er} is too useful to ignore and too impure to treat casually.

Recommended starting bounds and priors

These recommendations are methods-facing starting points, not final truths. They are best used as weakly informative bounds or prior anchors to be checked with posterior predictive diagnostics.

If the implementation uses $s = 1$

Parameter	Recommended starting region	Rationale
Drift rate v	Broad signed range, roughly -18 to $+18$, with most prior mass much closer to 0	Matches the outer empirical envelope once both response directions are allowed (Tran et al., 2020)
Boundary separation a	Roughly 0.2 to 3 as a practical region, with a harder plausibility ceiling near 7.5	Stays well inside the review-wide tail while respecting published fits (Tran et al., 2020)
Hand T_{er} for initial commitment	Roughly 0.15 to 0.35 s, with a softer center around 0.20 to 0.28 s	Tightened using reach preparation, eye-hand delay, and manual timing studies (Bompas et al., 2017; Bompas et al., 2024; Gopal et al., 2015; Haith et al., 2016)

Parameter	Recommended starting region	Rationale
Saccadic T_{er} for initial commitment	Roughly 0.07 to 0.15 s	Anchored by saccadic and manual-vs-saccadic timing work (Bompas et al., 2017; Bompas et al., 2024)

If the implementation uses $s = 0.1$

For v and a, divide the values above by 10. Time parameters are unchanged (Ratcliff et al., 2016; Tran et al., 2020).

Recommended modeling stance by condition

Condition	Best starting stance	Reason
Stationary-target reaching	Standard hierarchical DDM	Closest to a narrow initial commitment with less urgency and less need for online reinterpretation (Ratcliff & McKoon, 2008; Wiecki et al., 2013)
Moving-target interception	Hierarchical DDM as a baseline, interpreted cautiously	Urgency and online correction can cause fixed-bound parameters to absorb dynamics outside the standard model (Barany et al., 2020; Dendauw et al., 2024)
Joint eye-hand analysis	Shared decision stage plus effector-specific output terms where possible	Best fit with the coordinated eye-hand literature (Gopal et al., 2015; Jana et al., 2017)

Paper-by-paper backbone

Paper	Best use in the report	Why it matters
(Ratcliff & McKoon, 2008)	Core DDM theory	Defines what v, a, and T_{er} mean in the standard model
(Ratcliff & Tuerlinckx, 2002)	Fitting discipline	Supports contaminant handling and quantile-based assessment
(Vandekerckhove et al., 2011)	Hierarchical justification	Supports shrinkage and simultaneous group-plus-subject inference
(Wiecki et al., 2013)	Practical hierarchical implementation	Supports informative priors and HDDM-style modeling
(Tran et al., 2020)	Prior and bound source	Best empirical review for outer parameter envelopes

Paper	Best use in the report	Why it matters
(Ratcliff et al., 2016)	Scaling and interpretation caution	Prevents overclaiming about cross-study comparability
(Bompas et al., 2024)	Main T_{er} caution	Shows fitted nondecision time is often not literal
(Weindel et al., 2020)	Physiological decomposition	Shows motor time changes with nominally decisional manipulations
(Haith et al., 2016)	Hand timing floor	Gives a concrete preparation floor and initiation slack
(Bompas et al., 2017)	Eye versus hand timing	Supports separate output timing for eyes and hand
(Gopal et al., 2015)	Shared eye-hand decision stage	Supports common accumulation plus later effector-specific delay
(Jana et al., 2017)	Task-dependent architecture	Shows common versus separate accumulation depends on context
(Selen et al., 2012)	Continuous motor preparation	Supports overlap between deliberation and motor preparation
(Barany et al., 2020)	Interception-specific caution	Shows urgency and online redirection in a close task analogue
(Dendauw et al., 2024)	Extension beyond plain DDM	Explains how decision, preparation, and execution can overlap

Other papers worth considering

If the report needs a broader frame or the model becomes more ambitious, the next most useful papers are (Böhm et al., 2018) for across-trial variability, (Servant et al., 2021) and (Forstmann et al., 2015) for integrated decision–action framing, (Verdonck & Tuerlinckx, 2016) for nondecision-time separation, (Purcell et al., 2010; Purcell et al., 2012) for neural and oculomotor accumulator models, (Heitz & Schall, 2012) and (Cisek et al., 2009) for speed–accuracy and urgency, and (Gómez et al., 2015) for response-modality comparisons.

Final interpretation

The literature supports a careful but usable conclusion. A hierarchical DDM can be justified for Kinarm data if it is presented as a model of **initial commitment**, if priors on v and a are anchored to published DDM reviews under an explicit scaling convention, and if T_{er} is constrained using reaching, saccadic, and eye-hand evidence rather than treated as a free catch-all parameter (Bompas et al., 2024; Gopal et al., 2015; Haith et al., 2016; Jana et al., 2017; Tran et al., 2020).

The same literature also sets the limit of that claim. Moving-target interception and coordinated eye-hand behavior create urgency, overlap, and online correction that a plain fixed-bound diffusion model can summarize, but not fully explain (Barany et al., 2020; Dendauw et al., 2024; Selen et al., 2012). A good methods section should therefore justify the DDM clearly, use it pragmatically, and interpret it modestly.

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